

Advanced Topics in Computer Graphics II

Geometry Processing

Morphable Models



January 31, 2023





Problem: Given a large set of similar shape specimens. How to efficiently describe the shape variation for editing and fitting?



In this lecture: Focus on Faces.

Paper: V. Blanz and T. Vetter. "A morphable model for the synthesis of 3D faces". In: *Proceedings of the 26th annual conference on Computer graphics and interactive techniques*. 1999, pp. 187–194

But there are many other applications as well:

- ▶ Biology
- ▶ Medicine
- ▶ Image analysis

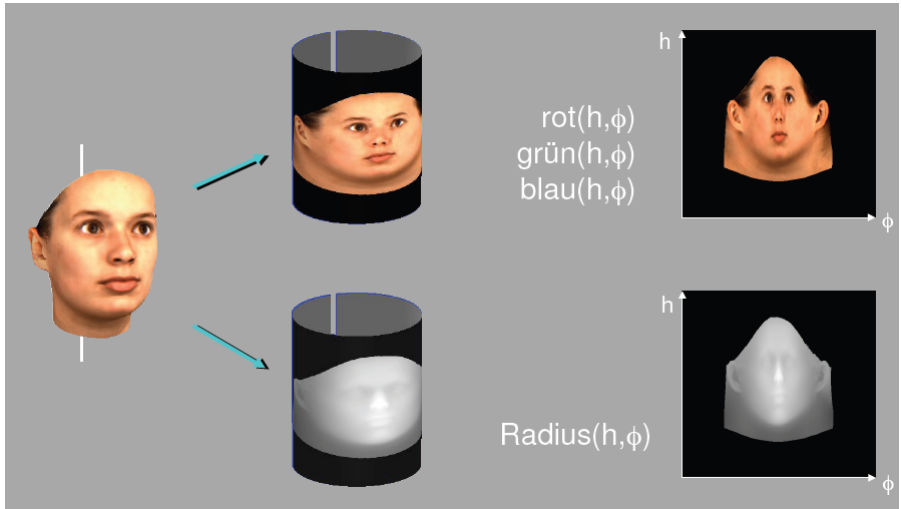


Database:

- ▶ 200 heads of young adults
- ▶ 100 males / 100 females
- ▶ Cylindrical representation, with radii $r(h, \phi)$ of surface points sampled at
 - ▶ 512 equally-spaced angles ϕ
 - ▶ 512 equally-spaced vertical steps h
- ▶ RGB color values per vertex
- ▶ Face is represented by 70,000 vertices
- ▶ Acquired with Cyberware Laserscanner (15 sec.)



A Morphable Model of Faces





Database





All m examples are in **full correspondence**

- ▶ This requires registration!

A face is given as

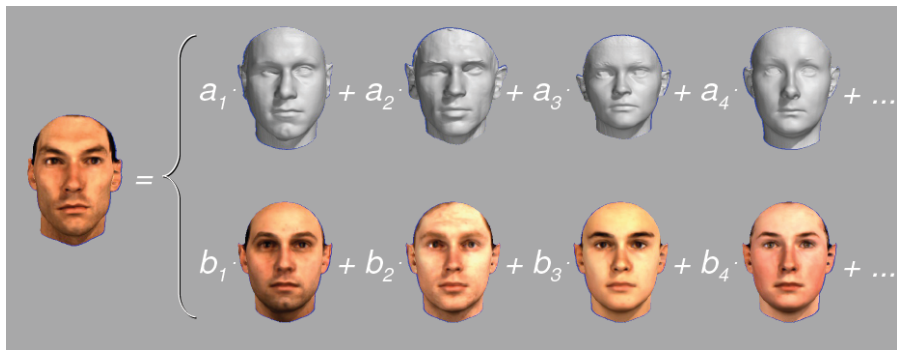
- ▶ $S_i = (X_1^i, Y_1^i, Z_1^i, \dots, X_N^i, Y_N^i, Z_N^i) \in \mathbb{R}^{3N}$
All vertex positions collected in a single vector
- ▶ $T_i = (R_1^i, G_1^i, B_1^i, \dots, R_N^i, G_N^i, B_N^i) \in \mathbb{R}^{3N}$
All vertex colors collected in a single vector



Observation: New shapes and textures can be expressed in barycentric coordinates as linear combinations of given specimens.

$$S_{new} = \sum_{i=1}^M a_i S_i \quad T_{new} = \sum_{i=1}^M b_i T_i$$

where $\sum_{i=1}^M a_i = 1$ and $\sum_{i=1}^M b_i = 1$



Morphable model: Set of faces $\{S_{new}(\mathbf{a}), T_{new}(\mathbf{b})\}$
Parameterized by coefficient vectors $\mathbf{a}$ and $\mathbf{b}$



Which coefficients yield plausible faces?

Assumption: Plausible faces are described by a multivariate normal distribution

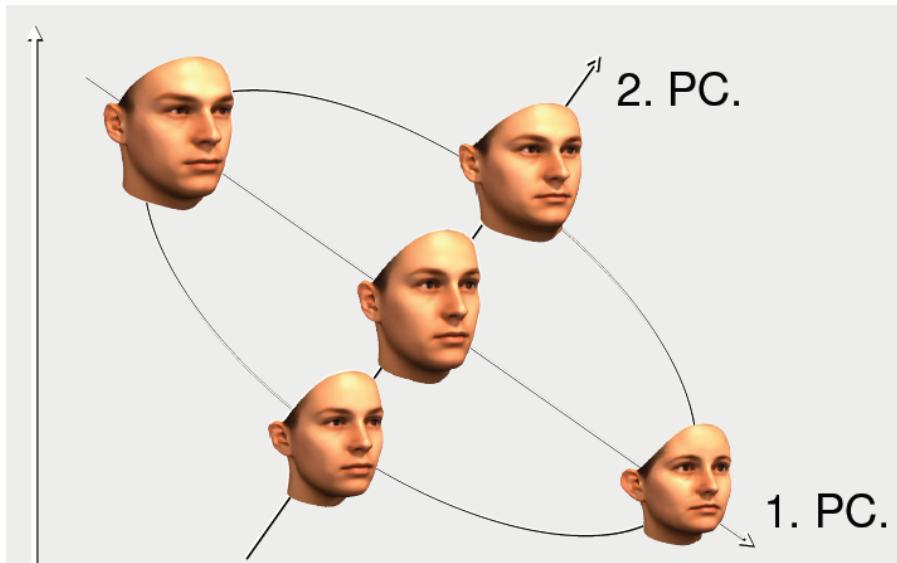
Distribution can be estimated using a PCA on S and T vectors:

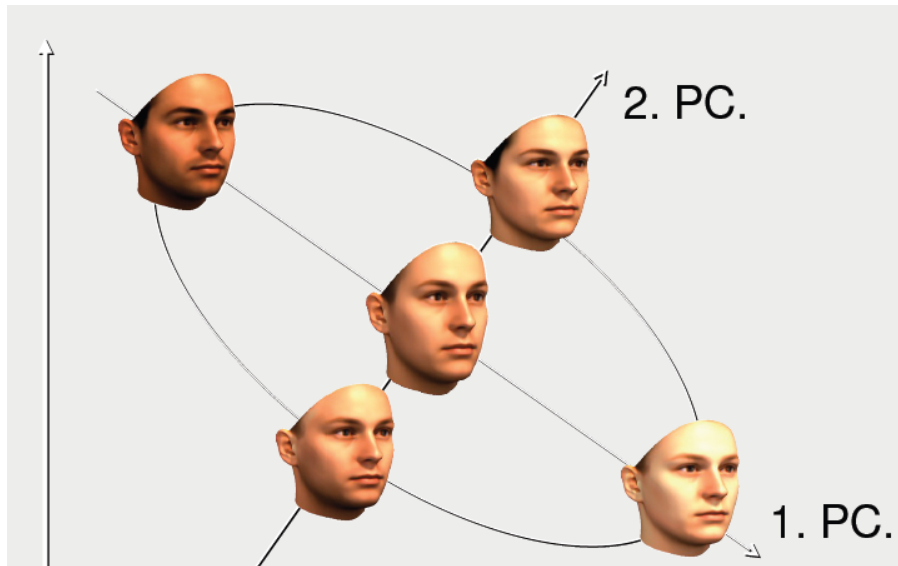
$$S_{new} = \bar{S} + \sum_{i=1}^{M-1} \alpha_i s_i \quad T_{new} = \bar{T} + \sum_{i=1}^{M-1} \beta_i t_i$$

where $\bar{S}$ and $\bar{T}$ are the mean vectors respectively and α_i and β_i are coefficients w.r.t. the principal components s_i and t_i .

The probability of a face is then given by

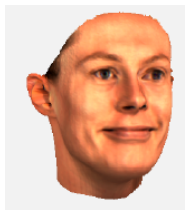
$$p(\alpha) = \exp\left(-\frac{1}{2} \sum_{i=1}^{M-1} \left(\frac{\alpha_i}{\sigma_i}\right)^2\right)$$







Two scans of the same individual with different expressions



-



= Smile

Transfer
pression

Ex-



+ Smile =





Transfer of Facial Expressions



original



surprise



joy



sadness



fear



anger



disgust



Facial expressions can be transferred from different examples of the same specimen.

What about other attributes such as

- ▶ gender
- ▶ darkness of eyebrows
- ▶ double chins
- ▶ etc.

where it is not possible to obtain different examples from the same individual?



Input: Set of faces $\{S_i, T_i\}$ manually assigned labels μ_i .

Labels describe importance of attribute in specimen.

Assumption: $\mu(S, T)$ is linear

$$\Delta S = \sum_{i=1}^M \mu_i (S_i - \bar{S}) \quad \Delta T = \sum_{i=1}^M \mu_i (T_i - \bar{T})$$

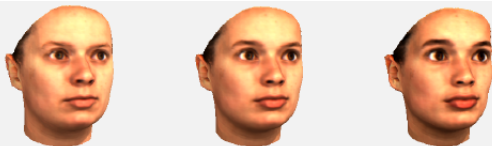
Define vectors for changing the attribute in a plausible manner w.r.t. the normal distribution assumption.



Facial Attributes



Attractiveness



Weight



Gender



original



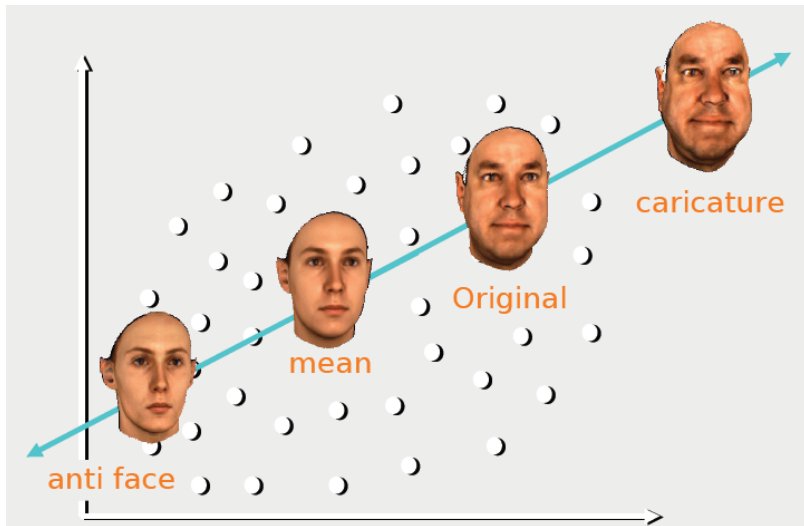
Male and Female Faces



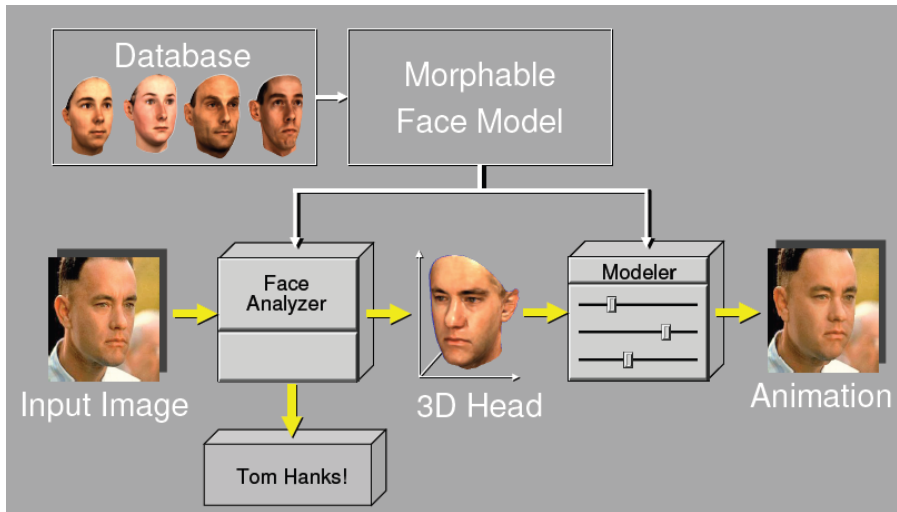
Morphing of gender

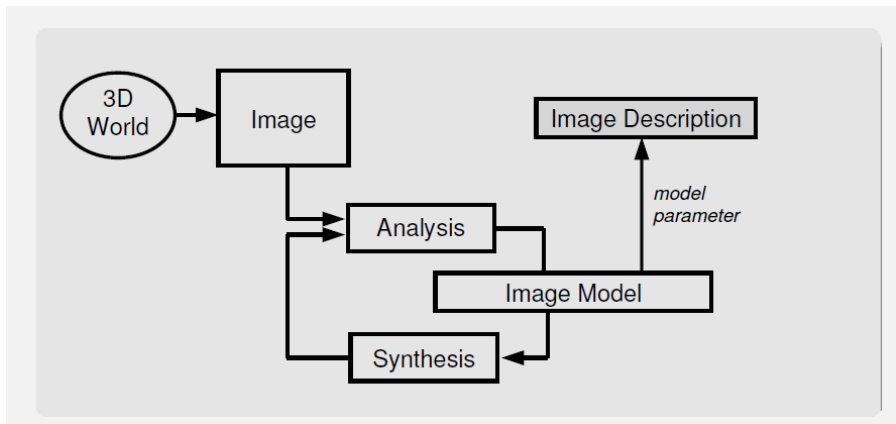


- Caricature by increasing the distance from the mean (extrapolation)
- Anti-face opposite face to original w.r.t. mean



Matching a Morphable Model to images





Analysis by synthesis loop: Renders an image and updates the parameters according to the residual difference



Parameters that need to be estimated:

- ▶ Shape coefficients: $\alpha_j, j \in \{1, \dots, M-1\}$
- ▶ Texture coefficients: $\beta_j, j \in \{1, \dots, M-1\}$
- ▶ Rendering parameters: ρ
 - ▶ Camera position (azimuth, elevation)
 - ▶ Object scale
 - ▶ Image plane rotation and translation
- ▶ Intensity of ambient and directed light (directly supplied by user): i_{amb}, i_{dir}
 - ▶ contrast,
 - ▶ offset
 - ▶ and gain



From $\{\alpha, \beta, \rho\}$ color images are rendered using

- ▶ Perspective projection
- ▶ Phong illumination model

The reconstructed image $I_{model} = I_{\{\alpha, \beta, \rho\}}$ is supposed to minimize the residual:

$$E_I = \sum_{x,y} \|I_{input}(x,y) - I_{model}(x,y)\|^2$$

Problem ill-posed:

- ▶ In general many non-face-like surfaces lead to the same image!

Therefore: Require the set of solutions to be plausible in the sense of the Morphable Model



Trade-off: Matching quality vs. prior probability

- Bayes decision theory: Find a set parameters with maximal posterior probability, given image I_{input}

Recall prior probability of face coefficients:

$$p(\alpha) = \exp\left(-\frac{1}{2} \sum_{i=1}^{M-1} \left(\frac{\alpha_i}{\sigma_i}\right)^2\right)$$

Likelihood to observe I_{input} if estimated parameters were correct (real images contain noise):

$$p(I_{input} | \alpha, \beta, \rho) = \exp\left(-\frac{E_I}{2\sigma^2}\right)$$



Overall approach: Minimize negative log-likelihood

$$E = \frac{1}{\sigma_N^2} E_I + \sum_{j=1}^{M-1} \frac{\alpha_j^2}{\sigma_{S,j}^2} + \sum_{j=1}^{M-1} \frac{\beta_j^2}{\sigma_{T,j}^2} + \sum_{j=1}^{M-1} \frac{(\rho_j - \bar{\rho}_j)^2}{\sigma_{\rho,j}^2}$$

where $\bar{\rho}_j, \sigma_{\rho,j}$ are user specified ad-hoc estimates



Simplification: Consider only centers of triangles because color can be easily evaluated there.

The pixel color in I_{model} at the projected location $(p_{x,k}, p_{y,k})$ of the center of triangle k is given by the Phong model as (for red component):

$$I_{r,model,k} = (i_{r,amb} + i_{r,dor} \cdot \langle n_k, l \rangle) \cdot r_k + i_{r,dir} \cdot s \cdot \langle r_k, v_k \rangle^\nu$$

where

- ▶ n_k normal of triangle
- ▶ l light direction
- ▶ s specularity
- ▶ r_k Reflected direction
- ▶ v_k viewing direction
- ▶ ν shininess



With this simplification we can write

$$E_I \approx \sum_{k=1}^{N_t} a_k \|I_{input}(p_{x,k}, p_{y,k}) - I_{model,k}\|^2$$

where a_k is the Image area of triangle k .

Further optimization: Select random subset of triangles $\mathcal{K} \subset \{1, \dots, N_t\}$

$$E_{k \in \mathcal{K}} = \sum_{k \in \mathcal{K}} a_k \|I_{input}(p_{x,k}, p_{y,k}) - I_{model,k}\|^2$$



From

$$E_{k \in \mathcal{K}} = \sum_{k \in \mathcal{K}} a_k \|I_{input}(p_{x,k}, p_{y,k}) - I_{model,k}\|^2$$

derivatives $\frac{\partial E_{\mathcal{K}}}{\partial \alpha_j}$, $\frac{\partial E_{\mathcal{K}}}{\partial \beta_j}$, $\frac{\partial E_{\mathcal{K}}}{\partial \rho_j}$ can be obtained.

Derivatives of texture and shape yield derivatives of $2D$ locations, surface normals, vectors ν_k and r_k and $I_{model,k}$.



Shadows and occlusion need to be handled:

- ▶ Render full model every 1000 iterations
 - ▶ Determine hidden triangles (depth-buffer) and exclude them from energy
 - ▶ Determine triangles in shadowed areas and adjust Phong-Model to use only ambient light for these triangles



Perform gradient descent with varying subset $\mathcal{K}$ every 1000 steps.

- ▶ Increase performance and resilience to local minima

Coarse-to-fine

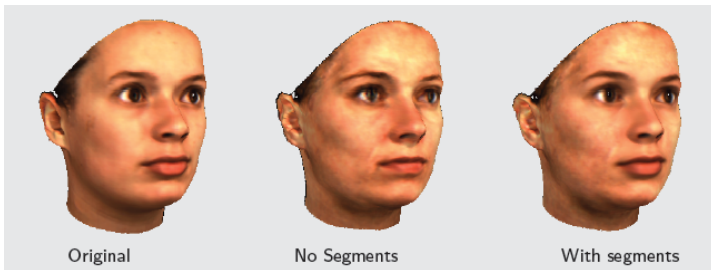
- ▶ First iterations on down-sampled input image
- ▶ Start only with the first α_i, β_i
- ▶ Start with large σ_N (strong weight on prior)
- ▶ In the last iterations: break the face model into segments



Segmented Morphable Model



- ▶ Define segments on reference face
- ▶ Match segments separately
- ▶ Blend results in 3D



Original

No Segments

With segments



Specific features are not captured by the Morphable Model (e.g. liver spot)

- ▶ Extract texture from image
- ▶ Separate pure albedo
 - ▶ Handle shading
 - ▶ Shadows
- ▶ After that: Able to change pose and illumination



2D input



rough initialization



automated 3D shape and texture reconstruction



illumination corrected
texture extraction



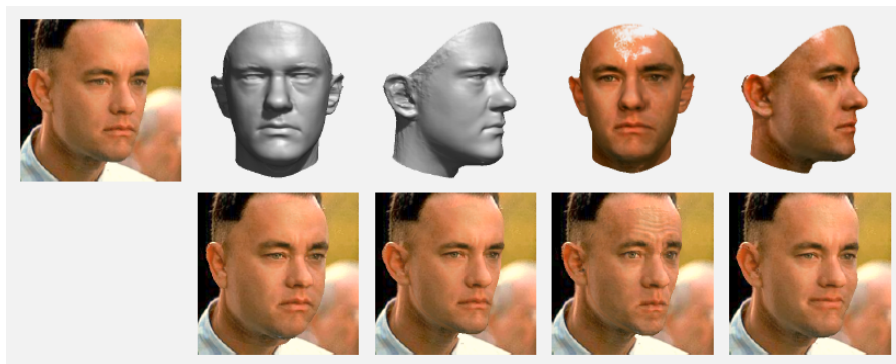
Audrey Hepburn



Audrey Hepburn (1929 - 1993) was an Academy Award-winning actress of film and theater, Broadway stage performer, ballerina, fashion model, and humanitarian.

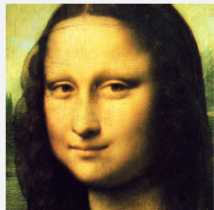


Tom Hanks





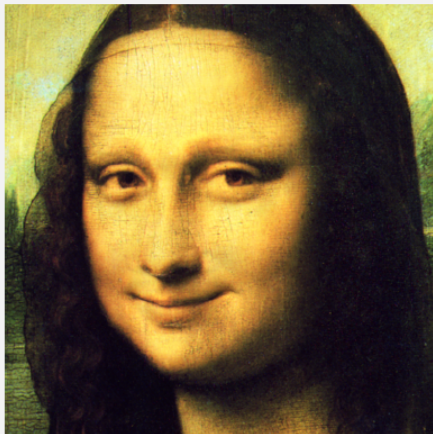
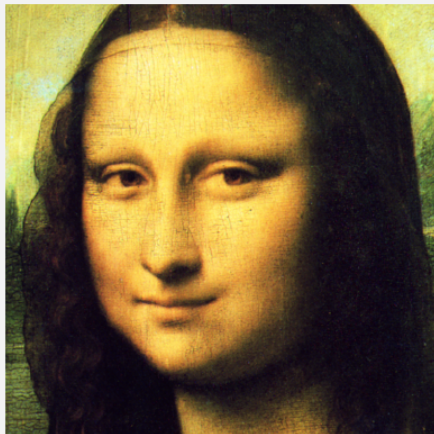
Mona Lisa



Mona Lisa, or La Gioconda. (La Joconde), is a 16th century oil painting on poplar wood by Leonardo da Vinci, and is one of the most famous paintings in the world. Few other works of art have been subject to as much scrutiny, study, mythologizing and parody. It is owned by the French government and hangs in the Musée du Louvre in Paris. The painting, a half-length portrait, depicts a woman whose gaze meets the viewer's with an expression often described as enigmatic. It is considered to be Leonardo's magnum opus.



Mona Lisa





- ▶ Partial occlusions (e.g. hair or glasses)
- ▶ Accuracy of shape reconstruction at contours
- ▶ Accuracy of illumination reconstruction (cast shadows)

Goal: Recover the 3D shape of human faces from a small set of points

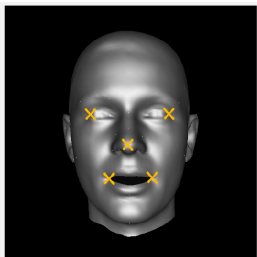
Assumption¹²: Class of object or data can be represented as a vector space spanned by a set of examples

¹V. Blanz and T. Vetter. "Reconstructing the complete 3D shape of faces from partial information (rekonstruktion der dreidimensionalen form von gesichtern aus partieller information)". In: *it-Information Technology* 44.6 (2002), pp. 295–302.

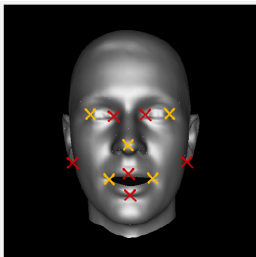
²V. Blanz, A. Mehl, et al. "A statistical method for robust 3D surface reconstruction from sparse data". In: *Proceedings. 2nd International Symposium on 3D Data Processing, Visualization and Transmission, 2004. 3DPVT 2004*. IEEE. 2004, pp. 293–300.



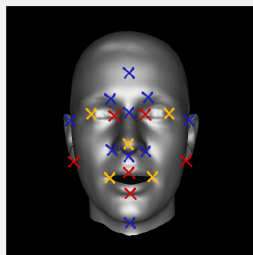
Landmark points



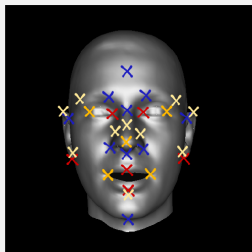
5 landmark points



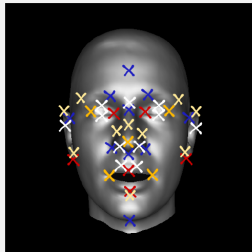
11 landmark points



21 landmark points



31 landmark points



42 landmark points

Recall: Given a set of example data vectors $\mathbf{v}_i$

- ▶ Mean-free data: $\mathbf{x}_i = \mathbf{v}_i - \bar{\mathbf{v}}$ with $\bar{\mathbf{v}} = \frac{1}{M} \sum_{i=1}^M \mathbf{v}_i$
- ▶ Data matrix: $X = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \dots & \mathbf{x}_m \\ | & | & & | \end{bmatrix}$
- ▶ (Unbiased) Covariance matrix: $\Sigma = \frac{1}{M-1} X X^T$ with eigendecomposition $\Sigma = S \cdot \Lambda \cdot S^T$ where $\Lambda = \text{diag}(\lambda_1^2, \dots, \lambda_m^2)$
- ▶ Use eigenvectors as basis: $\mathbf{x} = S \Lambda \cdot \mathbf{c}$ where $\mathbf{c}$ is the coefficient vector with c_i as coefficient from the projection of the data vector $\mathbf{x}$ onto the basis vectors $\mathbf{s}_i \lambda_i$

Assumption: Normal distribution in each of the eigendirections. These directions are uncorrelated!

Probability of a vector $\mathbf{x}$:

$$\begin{aligned} p(\mathbf{x}) &= \prod_{i=1}^M \frac{1}{\sqrt{2\pi}\lambda_i} \exp\left(-\frac{1}{2\lambda_i^2} \langle \mathbf{s}_i, \mathbf{x} \rangle^2\right) \\ &= \prod_{i=1}^M \frac{1}{\sqrt{2\pi}\lambda_i} \exp\left(-\frac{1}{2} c_i^2\right) \\ &= \left(\frac{1}{\sqrt{2\pi}}\right)^M \frac{1}{\prod_{i=1}^M \lambda_i} \exp\left(-\frac{1}{2} \sum_{i=1}^M c_i^2\right) \\ &= \frac{1}{(2\pi)^{M/2} \det(\Lambda)} \exp\left(-\frac{1}{2} \|\mathbf{c}\|^2\right) \end{aligned}$$



Problem: Given a measurement $\mathbf{r} \in \mathbb{R}^L$, $L < N$.

Find the full vector $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{r} = \mathcal{L}\mathbf{x}$ and a linear mapping $\mathcal{L}: \mathbb{R}^N \rightarrow \mathbb{R}^L$

- ▶ $\mathcal{L}$ can be any linear transformation, does not have to be a projection.
- ▶ We assume $\mathcal{L}$ is known.
- ▶ Admissible solutions should be in the span of the $\mathbf{x}_i$.

Usually it is not possible to find linear combinations of examples that solve directly

$$\mathbf{r} = \mathcal{L}\mathbf{x}$$

Therefore minimize

$$E(\mathbf{x}) = \|\mathcal{L}\mathbf{x} - \mathbf{r}\|^2$$



With $Q = \mathcal{L} \cdot S \cdot \Lambda \in \mathbb{R}^{L \times M}$

$$E(\mathbf{x}) = \|\mathcal{L}\mathbf{x} - \mathbf{r}\|^2$$

can be restated as

$$\begin{aligned} E(\mathbf{c}) &= \left\| \mathcal{L} \sum_{i=1}^M c_i \sigma_i \mathbf{s}_i - \mathbf{r} \right\|^2 \\ &= \left\| \sum_{i=1}^M c_i \mathbf{q}_i - \mathbf{r} \right\|^2 \\ &= \|Q\mathbf{c} - \mathbf{r}\|^2 \end{aligned}$$

Minimizing solution:

$$\mathbf{c}^* = Q^+ \mathbf{r}$$

where Q^+ is the pseudo inverse (computed via SVD).

The solution $\mathbf{c}^*$ is optimal in two respects:

1. $\mathbf{c}^*$ minimizes the energy E
2. Among all solutions $\{\mathbf{c}' | E(\mathbf{c}') = E(\mathbf{c})\}$, $\mathbf{c}$ has minimum norm $\|\mathbf{c}\|$



42 landmark points; 150 principal components;

Oops...



What went wrong?

- ▶ Always ensure that $E(c)$ is minimized
- ▶ Prior probability: withing solutions of equal $E(c)$
 - ▶ Measurement r cannot fully be reconstructed by elements of the object class
 - ▶ Noise
 - ▶ The full range of the object class is not covered by the set of examples

Minimizing $E(c)$ will lead to model coefficients far from the average (overfitting)!



Trade-off: Prior probability of the solution and matching quality

Assumption: Each dimension of the measured vector $\mathbf{r}$ is subject to uncorrelated Gaussian noise with variance σ_N^2 . With

$$\begin{aligned}\mathbf{r}_{model} &= \mathcal{L} \sum_{i=1}^M c_i \lambda_i \mathbf{s}_i \\ &= \sum_{i=1}^M c_i \mathbf{q}_i \\ &= Q\mathbf{c}\end{aligned}$$

$$\begin{aligned}p(\mathbf{r}|\mathbf{r}_{model}) &= \nu_N \cdot \exp\left(-\frac{1}{2\sigma_N^2} \|\mathbf{r}_{model} - \mathbf{r}\|^2\right) \\ &= \nu_N \cdot \exp\left(-\frac{1}{2\sigma_N^2} \|Q\mathbf{c} - \mathbf{r}\|^2\right)\end{aligned}$$



Bayes Theorem:

$$\begin{aligned}p(\mathbf{c}|\mathbf{r}) &= \nu p(\mathbf{r}|\mathbf{c})p(\mathbf{c}) \\&= \nu \cdot \nu_N \cdot \exp\left(-\frac{1}{2\sigma_N^2}\|Q\mathbf{c} - \mathbf{r}\|^2\right) \cdot \exp\left(-\frac{1}{2}\|\mathbf{c}\|^2\right) \\&= \nu \cdot \nu_N \cdot \exp\left(-\frac{1}{2\sigma_N^2}\|Q\mathbf{c} - \mathbf{r}\|^2 - \frac{1}{2}\|\mathbf{c}\|^2\right)\end{aligned}$$

This probability is maximized if the following cost function is minimized:

$$\begin{aligned}E(\mathbf{c}) &= -2 \cdot \ln p(\mathbf{c}|\mathbf{r}) \\&= \frac{1}{\sigma_N^2}\|Q\mathbf{c} - \mathbf{r}\|^2 + \|\mathbf{c}\|^2 + \text{const}\end{aligned}$$

This leads to the following energy, if $\eta = \sigma_N^2$

$$E(\mathbf{c}) = \|Q\mathbf{c} - \mathbf{r}\|^2 + \eta\|\mathbf{c}\|^2$$



$$\begin{aligned}E(\mathbf{c}) &= \|Q\mathbf{c} - \mathbf{r}\|^2 + \eta\|\mathbf{c}\|^2 \\&= \langle Q\mathbf{c}, Q\mathbf{c} \rangle - 2\langle Q\mathbf{c}, \mathbf{r} \rangle + \|\mathbf{r}\|^2 + \eta\|\mathbf{c}\|^2 \\&= \langle \mathbf{c}, Q^T Q\mathbf{c} \rangle - 2\langle \mathbf{c}, Q^T \mathbf{r} \rangle + \|\mathbf{r}\|^2 + \eta\|\mathbf{c}\|^2\end{aligned}$$

In the optimum:

$$\begin{aligned}0 &= \nabla E(\mathbf{c}) \\&= 2Q^T Q\mathbf{c} - 2Q^T \mathbf{r} + 2\eta\mathbf{c}\end{aligned}$$

With $Q = UWV^T$

$$\begin{aligned}VW^2V^T\mathbf{c} + \eta\mathbf{c} &= VWU^T\mathbf{r} \\V(W^2 + \eta)V^T\mathbf{c} &= VWU^T\mathbf{r} \\ \mathbf{c} &= V \operatorname{diag}\left(\frac{w_i}{w_i^2 + \eta}\right) U^T \mathbf{r}\end{aligned}$$

where w_i are the singular values stored in the matrix W .



original

42 landmark points
150 principal components



reconstruction $\eta = 10^0$

$d_{\text{fullshape}} = 64.20\text{ mm}$
 $d_{\text{landmarks}} = 0.07\text{ mm}$

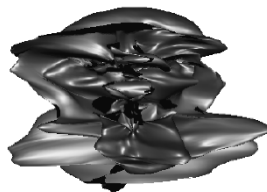


Results



original

42 landmark points
150 principal components



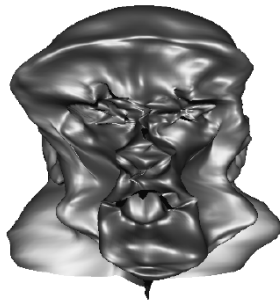
reconstruction $\eta = 10^1$

$d_{\text{fullshape}} = 32.47\text{mm}$
 $d_{\text{landmarks}} = 0.14\text{mm}$



original

42 landmark points
150 principal components



reconstruction $\eta = 10^2$

$d_{\text{fullshape}} = 15.06\text{mm}$
 $d_{\text{landmarks}} = 0.23\text{mm}$



original

42 landmark points
150 principal components



reconstruction $\eta = 10^3$

$d_{\text{fullshape}} = 7.00\text{mm}$
 $d_{\text{landmarks}} = 0.33\text{mm}$



original

42 landmark points
150 principal components



reconstruction $\eta = 10^4$

$d_{\text{fullshape}} = 3.84\text{mm}$
 $d_{\text{landmarks}} = 0.54\text{mm}$



original

42 landmark points
150 principal components



reconstruction $\eta = 10^5$

$d_{\text{fullshape}} = 2.95\text{mm}$
 $d_{\text{landmarks}} = 0.94\text{mm}$



original

42 landmark points
150 principal components



reconstruction $\eta = 10^7$

$d_{\text{fullshape}} = 2.49\text{mm}$
 $d_{\text{landmarks}} = 2.45\text{mm}$



original

42 landmark points
150 principal components

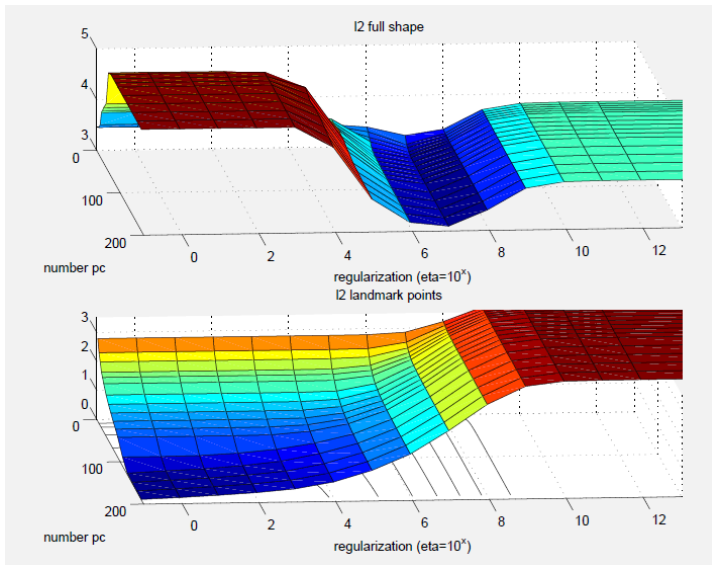


reconstruction $\eta = 10^8$

$d_{\text{fullshape}} = 3.21\text{mm}$
 $d_{\text{landmarks}} = 3.34\text{mm}$



Results: 42 Landmarks





Key problem: Compute dense point-to-point correspondences between the vertices of all the faces

Compute correspondences with optical-flow!

Given two cylindrical scans $I_1(h, \phi)$ and $I_2(h, \phi)$ optical-flow computes a flow-field $(\delta h(h, \phi), \delta \phi(h, \phi))$ such that $\|I_1(h, \phi) - I_2(h - \delta h, \phi + \delta \phi)\|$ is minimized (equally weighting shape and texture).



Because the optical-flow is unconstrained and there exist facial regions with little structure, the flow-field needs to be smoothed.

After the smoothing the optical-flow is recomputed using the smoothed field as initialization. This is iterated on increasing levels of resolution.



Optical-flow can be used to compute correspondences between two scans. But all scans are needed in correspondence.

Employ simple bootstrapping procedure!

1. Select an arbitrary face as initial reference face.
2. Compute correspondences of all faces to the reference.
3. Use these correspondences to compute initial Morphable Model.
4. Match current Morphable Model to all faces using most significant components.
5. Use optical-flow to find correspondence between matched model and each face.
6. Iterate and increase number of components.

PCA-based approach

- ▶ Example shapes

$$\{\Gamma_1, \dots, \Gamma_n\}$$

- ▶ Individual shapes with N registered landmark points

$$\Gamma_i = \{\mathbf{x}_k^i | \mathbf{x}_k \in \mathbb{R}^3, \quad k = 1, \dots, N\}$$

each represented by a vector

$$\mathbf{s}_i = (x_{1x}^i, x_{1y}^i, x_{1z}^i, \dots, x_{Nx}^i, x_{Ny}^i, x_{Nz}^i) \in \mathbb{R}^3$$

- ▶ Assumption (normal distribution)

$$\mathbf{s}_i \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

- ▶ With shape mean

$$\boldsymbol{\mu} = \bar{\mathbf{s}} := \frac{1}{n} \sum_{i=1}^n \mathbf{s}_i$$

- ▶ and shape covariance

$$\boldsymbol{\Sigma} = \mathbf{S} := \frac{1}{n-1} \sum_{i=1}^n (\mathbf{s}_i - \bar{\mathbf{s}})(\mathbf{s}_i - \bar{\mathbf{s}})^T$$

Problem: N is usually large, i.e. the covariance matrix Σ cannot be represented explicitly

- ▶ Fortunately Σ is determined by the n example datasets and therefore has at most rank n and can be represented using at most n basis vectors
- ▶ Probabilistic interpretation

$$\mathbf{s} = \bar{\mathbf{s}} + \sum_{i=1}^n \alpha_i \sqrt{\lambda_i} \mathbf{v}_i \sim \mathcal{N}(0, 1)$$

where $(\lambda_i, \mathbf{v}_i)$, $i = 1, \dots, n$ are the eigenvectors and eigenvalues of the covariance matrix Σ



Gaussian process morphable models: different interpretation of

$$\mathbf{s} = \bar{\mathbf{s}} + \sum_{i=1}^n \alpha_i \sqrt{\lambda_i} \mathbf{v}_i \sim \mathcal{N}(0, 1)$$

as model of deformations

$$\mathbf{u} = \sum_{i=1}^n \alpha_i \sqrt{\lambda_i} \mathbf{v}_i \sim \mathcal{N}(0, \mathbf{S})$$

with which the mean shape $\bar{\mathbf{s}}$ is deformed.

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4

³M. Lüthi, A. Forster, et al. “Shape modeling using gaussian process morphable models”. In: *Statistical Shape and Deformation Analysis*. Elsevier, 2017, pp. 165–191.

⁴M. Lüthi, T. Gerig, et al. “Gaussian process morphable models”. In: *IEEE transactions on pattern analysis and machine intelligence* 40.8 (2017), pp. 1860–1873.



A continuous stochastic process

$$\{X_t | t \in T\}$$

is Gaussian if and only if for every finite set of indices $t_1, \dots, t_k$ in the index set T

$$X_{t_1, \dots, t_k} = (x_{t_1}, \dots, x_{t_k})$$

is a multivariate Gaussian random variable, i.e. every linear combination

$$Y = a_1 X_{t_1} + \dots + a_k X_{t_k}$$

has a univariate normal distribution.

Gaussian processes are completely defined by their second order statistics.

If a Gaussian process is assumed to have mean zero, defining the covariance function completely defines the process' behavior.

Let $\Gamma_R \subset \mathcal{R}^3$ be a reference shape and $\Omega \subset \mathcal{R}^3$ a domain with $\Gamma_R \subset \Omega$

$$\mu: \Omega \rightarrow \mathbb{R}^3$$

a mean function and

$$k: \Omega \times \Omega \rightarrow \mathbb{R}^{3 \times 3}$$

a covariance function.

Then any deformation sampled from the corresponding Gaussian process $GP(\mu, k)$ defines a new shape

$$\Gamma = \{x + \hat{\mathbf{u}}(x) | x \in \Gamma_R\}$$



Similar to the PCA representation of a statistical shape model a Gaussian process $GP(\mu, k)$ can be represented in terms of an orthogonal set of basis functions $\{\phi_i\}_{i=1}^{\infty}$ (**Karhunen-Loeve expansion**)

$$\mathbf{u}(x) \sim \mu(x) + \sum_{i=1}^{\infty} \alpha_i \sqrt{\lambda_i} \phi_i, \quad \alpha_i \sim \mathcal{N}(0, 1)$$

where (λ_i, ϕ_i) are the eigenvalue/eigenfunction pairs of the integral operator

$$T_j f(\cdot) := \int_{\Omega} k(x, \cdot) f(x) d\rho(x)$$

where $\rho(x)$ denotes a measure on $\Omega \subset \mathcal{R}^3$.



The **Karhunen-Loeve expansion** can only be computed if the eigenfunction/eigenvalue pairs (λ_i, ϕ_i) are known.

Note that in applications a low-rank approximation using only r components of the expansion is sufficient.

Nyström method: Estimate eigenfunctions/eigenvalues of

$$T_k f(\cdot) := \int_{\Omega} k(x, \cdot) f(x) d\rho(x),$$

i.e.

$$\lambda_i \phi_i(x') = \int_{\Omega} k(x, x') \phi_i(x) d\rho(x), \quad \forall x' \in \Omega$$



$$\lambda_i \phi_i(x') = \int_{\Omega} k(x, x') \phi_i(x) d\rho(x), \quad \forall x' \in \Omega$$

Let $d\rho(x) = p(x)dx$ with density distribution. Generating samples $(x_l)_{l=1, \dots, n}$ according to $p(x)$ and using these samples for x' lead to a matrix problem

$$\mathbf{K} \mathbf{u}_i = \lambda_i^{mat} \mathbf{u}_i$$

where

- ▶ $\mathbf{K}_{il} = k(x_i, x_l)$ is the kernel matrix
- ▶ $\mathbf{u}_i$ is the i th eigenvector and
- ▶ λ_i^{mat} the corresponding eigenvalue



Advantages:

- ▶ Much more freedom in defining the covariance function than in classical model.
- ▶ Expressive prior models for registration can be derived.
- ▶ Arbitrary kernel functions can be used to define the covariance matrix.
- ▶ Combining different covariance functions (or kernels) allows to mimic more sophisticated registration schemes.



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